

PAPER_197: F_U_Bi_i Extended Integral — UV, mm-Wave, Hybrid, and Hierarchical Terms

Version: 1.0

Date: March 13, 2026

Session: 50 — grok_share_7514fe.txt Full Audit

Author: Star-Magic UQFF Research Framework

Source: grok_share_7514fe.txt lines 200-400

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

Abstract

This paper documents the extended form of the F_U_Bi_i buoyancy integral, incorporating four new terms beyond the standard UQFF formulation: F_UV (GALEX/Spitzer UV flare coupling), F_mm (ALMA mm-radio coupling), F_hyb (hybrid polarization-frequency term), and F_hier (hierarchical remnant unification). Numerical coefficients k_UV = k_mm = 10³⁰ N/W and f_mm = 1.05 are derived from observational calibration. This extended integral enables multi-wavelength coupling of buoyancy forces from UV through millimeter radiation fields.

1. Standard F_U_Bi_i Integral (Baseline)

The standard buoyancy integral form:

$$F_{U_Bi} = -F_0 + (m_e c^2/r^2) \cdot DPM_momentum \cdot \cos\theta + (GM/r^2) \cdot DPM_gravity + F_{U_Bi_i}$$

2. Extended F_U_Bi_i Integral

The complete extended form including all observational coupling terms:

$$F_{U_Bi_i} = \theta^{\{x2\}} [\\ -F_0 \\ + (m_e c^2/r^2) \cdot DPM_mom \cdot \cos\theta \\ + (GM/r^2) \cdot DPM_grav \\ + \theta_vac, [UA] \cdot DPM_stab \\ + k_LENR \cdot (\theta_LENR/\theta)^2 \\ + k_act \cdot \cos(\theta_act \cdot t) \\ + k_DE \cdot L_X \\ + 2qB_0V \cdot \sin\theta \cdot DPM_res \cdot P_pol \\ + k_neutron \cdot s_n \\ + k_rel \cdot (E_cm,astro,enhanced/E_cm)^2 \\ + k_UV \cdot L_UV \quad \quad \quad ? \text{ NEW: UV flare coupling} \\ + k_mm \cdot L_mm \cdot f_mm \quad \quad ? \text{ NEW: mm-radio coupling} \\] dx$$

3. New Extended Terms

3.1 F_{UV} — GALEX/Spitzer UV Coupling

$$F_{UV} = k_{UV} \cdot L_{UV}$$

Parameters:

$k_{UV} = 10^{30}$ N/W (UV force coupling constant)

L_{UV} = UV luminosity (W) from GALEX or Spitzer photometry

Physical origin: UV flare irradiation pressure on buoyant plasma shells

3.2 F_{mm} — ALMA mm-Radio Coupling

$$F_{mm} = k_{mm} \cdot L_{mm} \cdot f_{mm}$$

Parameters:

$k_{mm} = 10^{30}$ N/W (mm-radio force coupling constant)

L_{mm} = mm-radio luminosity (W) from ALMA observations

$f_{mm} = 1.05$ (mm-radio frequency enhancement factor)

Physical origin: millimeter-wave radiation pressure in molecular cloud outflows

3.3 F_{hyb} — Hybrid Polarization-Frequency Term

$$F_{hyb} = P_{pol} \cdot f_{mm} \cdot (\omega_0)^{-1}$$

Parameters:

P_{pol} = polarization fraction (dimensionless, typically 0.01–0.1)

$f_{mm} = 1.05$

ω_0 = base UQFF angular frequency (rad/s)

Physical origin: coupling of polarized mm-emission to buoyancy oscillation frequency

3.4 F_{hier} — Hierarchical Remnant Unification

$$F_{hier} = S \cdot (v_i/c)^n \cdot \omega_0^{-m}$$

Parameters:

v_i = velocity of remnant component i (m/s)

c = speed of light

$n = 2$ (hierarchical power index)

$m = 1$ (frequency suppression exponent)

Physical origin: multi-component remnant velocity hierarchy (e.g., jet + cocoon + lobe)

4. Standard Integral Terms (Reference)

Term	Symbol	Physical Origin
Base restoring force	$-F_0$	Vacuum restoring force
DPM momentum	$(\frac{m_e}{c^2/r^2}) \cdot DPM_{mom} \cdot \cos\theta$	Dipole-plasma momentum scattering
DPM gravity	$(GM/r^2) \cdot DPM_{grav}$	Dipole-plasma gravitational coupling

Term	Symbol	Physical Origin
DPM stability	$\varphi_{\text{vac}}[\text{UA}] \cdot \text{DPM_stab}$	Vacuum aether stability term
LENR coupling	$k_{\text{LENR}} \cdot (\varphi_{\text{LENR}}/\varphi_0)^2$	Low-energy nuclear resonance
Activation term	$k_{\text{act}} \cdot \cos(\varphi_{\text{act}} \cdot t)$	Activation oscillation
Dark energy luminosity	$k_{\text{DE}} \cdot L_X$	X-ray dark energy coupling
DPM resonance	$2qB_0V \cdot \sin\varphi \cdot \text{DPM_res} \cdot P_{\text{pol}}$	Magnetic resonance polarization
Neutron cross-section	$k_{\text{neutron}} \cdot s_n$	Neutron scattering
Relativistic CM	$k_{\text{rel}} \cdot (E_{\text{cm,astro,enh}}/E_{\text{rel}})^2$	Relativistic center-of-mass enhancement

5. Numerical Results for Extended Terms

System	Standard $F_{\text{U_Bi_i}}$	With UV/mm extension
Magnetar SGR1745	$\sim 2.11 \times 10^{20} \text{ N}$	+ F_{UV} from Chandra/GALEX
NGC 3603	$\sim -8.31 \times 10^{211} \text{ N}$	+ F_{mm} from ALMA CO observations
Pillars of Creation	$\sim 9.79 \times 10^{33} \text{ N}$	+ F_{hyb} with $P_{\text{pol}} \sim 0.05$

Note: The extreme magnitudes (10^{20} N , 10^{211} N) reflect vacuum-density-scaled units in the UQFF framework where $\varphi_{\text{vac}}[\text{UA}] \sim 10^{113}$ (dimensionless normalized).

6. Integration with Existing UQFF Terms

The extended integral fits within the Triadic Master System (PAPER_196) as the primary buoyancy channel $F_{\text{U_Bi}}$. Specifically:

- **F_{UV}** activates when GALEX UV flux > threshold (flare events)
- **F_{mm}** activates when ALMA observes SFR-driven mm continuum
- **F_{hyb}** operates continuously as polarization coupling
- **F_{hier}** applies to multi-component systems (AGN jets, SNR shells)

7. References

- grok_share_7514fe.txt lines 200–400 (first PDF extended integral section)
- PAPER_196: Triadic Master Equation System
- PAPER_171: Universal Gravity U_{g1} – U_{g4} Decomposition
- PAPER_172: FU Complete Unified Field Assembly